



Supplementary Materials for

A human homolog of SIR2 antiphage proteins mediates immunity via the Toll-like receptor pathway

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Tables S1 to S4

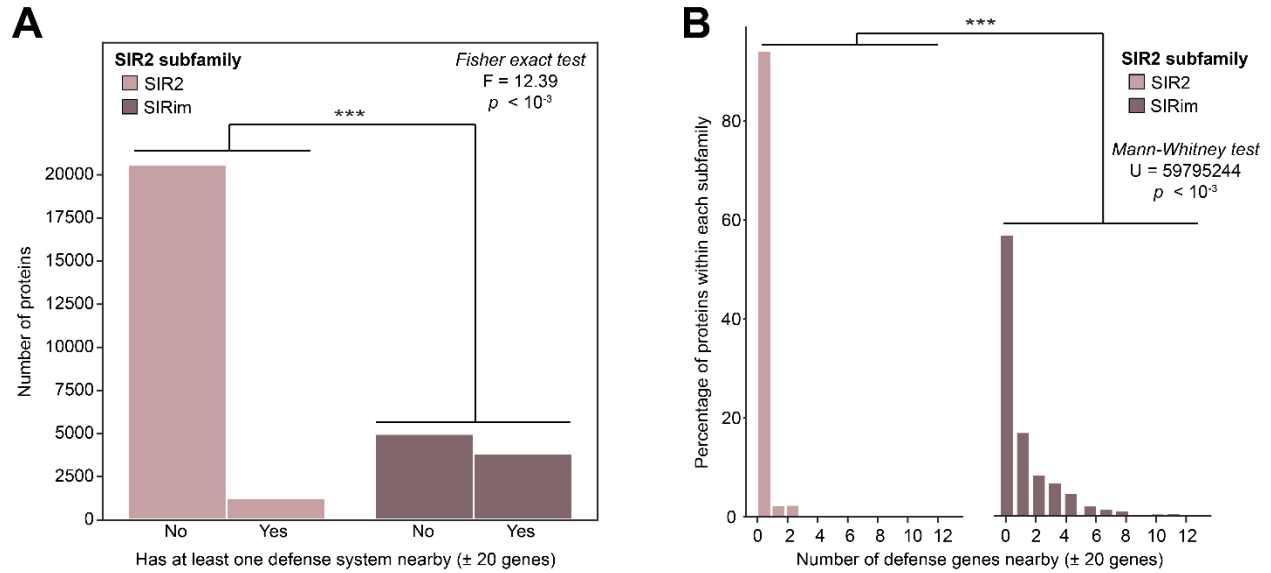
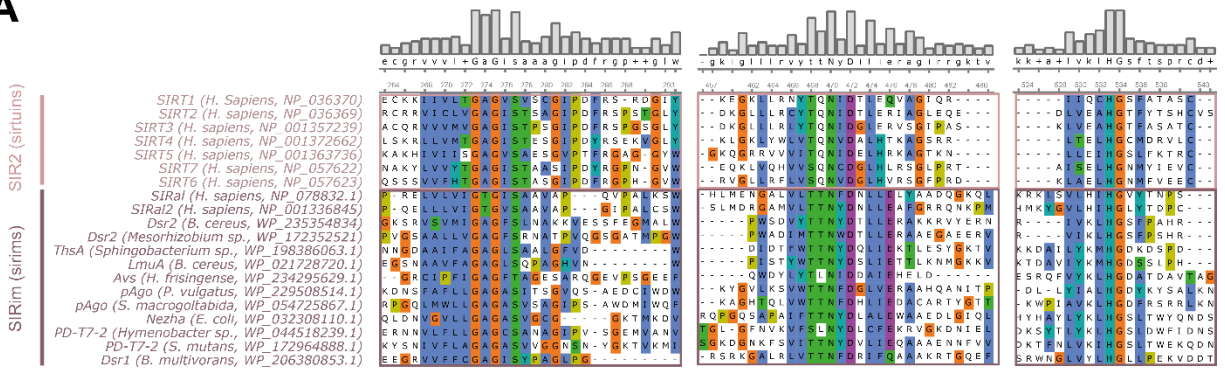
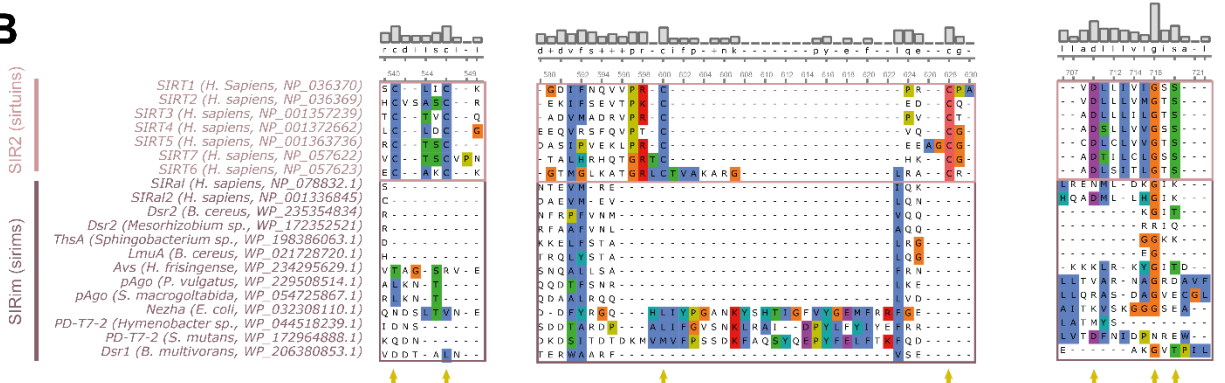


Fig. S1. The genomic context around bacterial proteins from the SIRim subfamily is significantly enriched in known antiphage systems compared with the genomic context nearby bacterial SIR2 subfamily proteins.

(A) Number of bacterial proteins which have at least one defense system in their genomic neighborhood (20 genes upstream and 20 genes downstream). Proteins are grouped according to their SIR2 subfamily: SIR2 (beige) and SIRim (brown). Two-sided Fisher exact test is used to assess statistical significance of the difference between the two distributions (***: $p < 10^{-3}$).

(B) Distribution of the number of defense genes in the genomic context of bacterial SIR2 family proteins per subfamily: SIR2 (beige) and SIRim (brown). The x-axis (“Number of defense genes nearby”) is duplicated across the two subfamilies for visualization purposes, but the y-axis (“Percentage of proteins within each subfamily”) is shared. Two-sided Mann-Whitney test is used to assess statistical significance of the difference between the two distributions (***: $p < 10^{-3}$).

A**B****Conserved residues**

→ Conserved across the whole SIR2 family (SIR2 and SIRim subfamilies)

→ Conserved within a single subfamily (SIR2 or SIRim subfamilies)

Fig. S2. Multiple sequence alignment of twenty SIR2 family proteins sampled from bacteria and eukaryotes.

Sequences are grouped by subfamily (SIR2: beige box, SIRim: brown box). Conserved residues are indicated by an arrow. **(A)** Selected regions of the multiple sequence alignment which exhibit strong conservation both within subfamilies and among subfamilies. All the conserved motifs (G.G, ..N.D and .HG) are located within the NAD⁺ binding site of the proteins (12, 17, 20, 22). **(B)** Selected regions of the multiple sequence alignment which exhibit strong conservation within one of the two subfamilies only. The four cysteine (C) residues conserved within the SIR2 subfamily (beige) form a tetrad which coordinates a Zn²⁺ ion and forms the binding site of deacetylated amino acid residue (Fig. 1B) (22). This tetrad is absent from all SIRim proteins (brown).

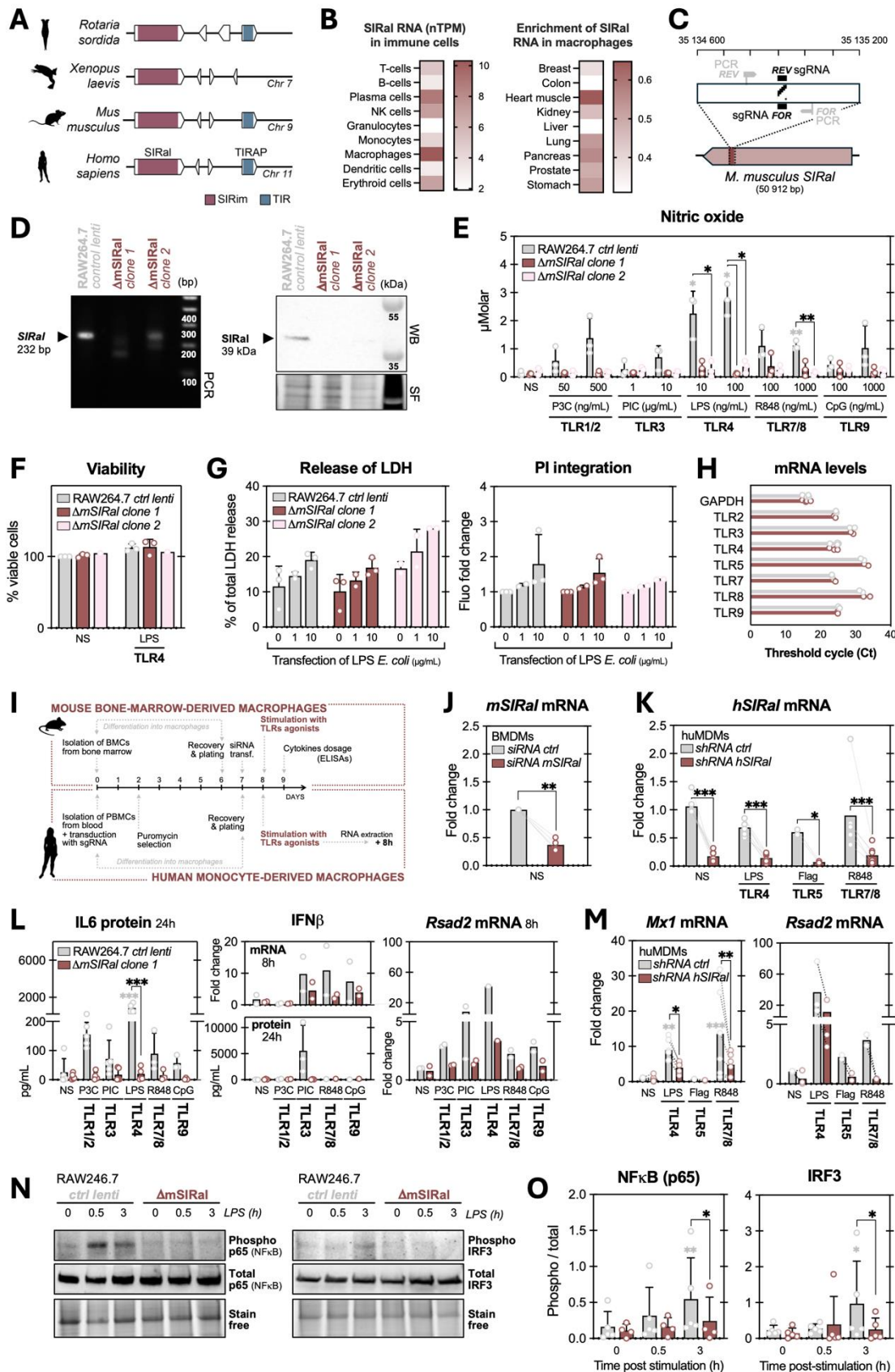


Fig. S3. Characterization of the role of *SIRal* in the TLR pathway in macrophages.

(A) Examples of *SIRal* genomic loci in animal genomes (metazoa). *SIRal* (pink) colocalizes with *TIRAP* (blue) in 73% of metazoa. In *Xenopus laevis*, the African clawed frog, *TIRAP* and *SIRal* are located on the same chromosome, but encoded at different loci. **(B)** Heat maps of *SIRal* RNA levels (number of transcripts per million, nTPM) in immune cells and enrichment ratio of *SIRal* RNA in macrophages of various tissues (Genotype-Tissue Expression, GTEx, Broad Institute). **(C)** Knock-out strategy for *mSIRal* in mouse RAW264.7 cells. **(D)** PCR on genomic DNA of wild type or $\Delta mSIRal$ (two clones) RAW264.7 cells (left). Western blot & stain free (WB & SF) of wild type or $\Delta mSIRal$ (two clones) RAW264.7 cells (right). **(E)** Wild type or $\Delta mSIRal$ (two clones) RAW264.7 cells were stimulated with Pam3CysSK4 (500ng/mL), Poly(I:C) (1 μ g/mL), LPS from *E. coli* (100ng/mL), R848 (1 μ g/mL) or CpG (1 μ g/mL), agonists of TLR1/2, TLR3, TLR4, TLR7/8 and TLR9, respectively. Production of nitric oxide (NO) was measured at 24h. **(F)** Wild type or $\Delta mSIRal$ (two clones) RAW264.7 cells were stimulated with LPS from *E. coli* (100ng/mL). Viability was measured by Alamar Blue assay at 24h. **(G)** Wild type or $\Delta mSIRal$ (two clones) RAW264.7 cells were transfected with LPS from *E. coli* (1-10 μ g/mL). Release of lactate dehydrogenase (LDH) and propidium iodide (PI) integration were measured at 24h. **(H)** Levels of *GAPDH* and *TLR* transcripts in wild type or $\Delta mSIRal$ RAW264.7 cells measured by RT-qPCR. **(I)** Experimental setup using mouse bone-marrow derived macrophages (BMDMs) and human monocyte-derived macrophages (huMDMs) to probe the activation of the TLR pathway. **(J)** Levels of *mSIRal* mRNA measured by RT-qPCR in mouse BMDMs transfected with a control siRNA, or a siRNA targeting *mSIRal*. **(K)** Levels of *hSIRal* mRNA measured by RT-qPCR in huMDMs transduced with a control shRNA, or a shRNA targeting *hSIRal* and stimulated with the indicated TLR agonists. **(L)** Wild type or $\Delta mSIRal$ RAW264.7 cells were stimulated with Pam3CysSK4 (500ng/mL), Poly(I:C) (1 μ g/mL), LPS from *E. coli* (100ng/mL), R848 (1 μ g/mL) or CpG (1 μ g/mL), agonists of TLR1/2, TLR3, TLR4, TLR7/8 and TLR9, respectively. Production of IL6 and IFN β was measured at 24h. Levels of *IFN β* and *Rsad2* transcripts were measured by RT-qPCR at 8h. **(M)** huMDMs transduced with a control shRNA, or a shRNA targeting *hSIRal*, were stimulated with LPS (100ng/mL), flagelin (500ng/mL) and R848 (1 μ g/mL), agonists of TLR4, TLR5 and TLR7/8, respectively. Levels of *Mx1* and *Rsad2* transcripts were measured by RT-qPCR at 8h. **(N, O)** Wild type or $\Delta mSIRal$ RAW264.7 cells were stimulated with LPS from *E. coli* (100ng/mL). Both total and phosphorylated p65 (NF- κ B) and IRF3 were monitored by Western blots (N) at 30min and 3h and quantification of the (phosphorylated/total) ratio was performed (O).

Each dot represents the mean of technical replicates for each independent experiment (E, F, G, H, L, O), independent mouse BMDMs isolation (J) or independent human donor (K, M). Bars correspond to the mean and error bars to the standard deviation of all the independent experiments/mouse/donors. Statistical analyses were performed using two-ways ANOVA and p-values were classified as * p<0.05; ** p<0.01; *** p<0.001. Grey stars correspond to comparisons within one group and dark stars correspond to comparisons between groups.

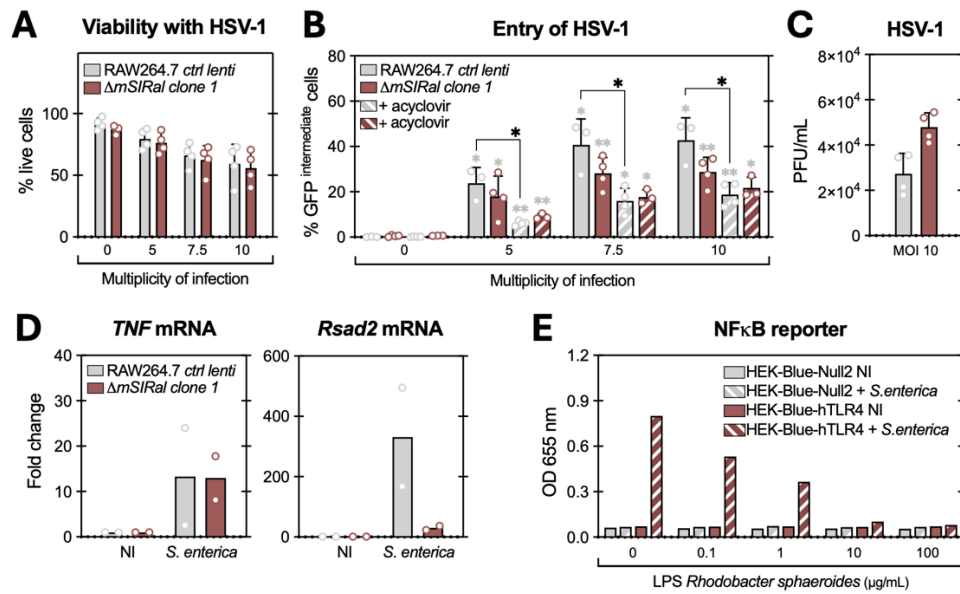


Fig. S4. SIRal participates in TLR-driven control of infection.

(A) Wild type or $\Delta mSIRal$ RAW264.7 cells were infected or not with HSV-1 (MOI 5-10). Viability was measured by flow cytometry at 24h. (B) Wild type or $\Delta mSIRal$ RAW264.7 cells were infected or not with HSV-1 (MOI 5-10), in the presence or absence of acyclovir (dashed). Levels of GFP were measured by flow cytometry at 24h and the percentage of entry (GFP^{intermediate}) was determined. (C) Wild type or $\Delta mSIRal$ RAW264.7 cells were infected or not with HSV-1 (MOI 10). Viral loads in the supernatant were quantified by plaque assay at 24h. (D) Wild type or $\Delta mSIRal$ RAW264.7 cells were infected with *Salmonella enterica* (MOI 10). Levels of *TNF* and *Rsad2* transcripts were measured by RT-qPCR at 9h. (E) HEK293T-hTLR4 NF-κB reporter cells were stimulated with PFA-fixed *Salmonella enterica*, in the presence or absence of LPS from *Rhodobacter sphaeroides* (1μg/mL), used as a TLR4 antagonist. NF-κB activation was measured at 24h.

Each dot represents the mean of technical replicates for each independent experiment (A, B, D). Bars correspond to the mean and error bars to the standard deviation of all the independent experiments. Statistical analyses were performed using two-ways ANOVA and p-values were classified as * p<0.05; ** p<0.01; *** p<0.001. Grey stars correspond to comparisons within one group and dark stars correspond to comparisons between groups.

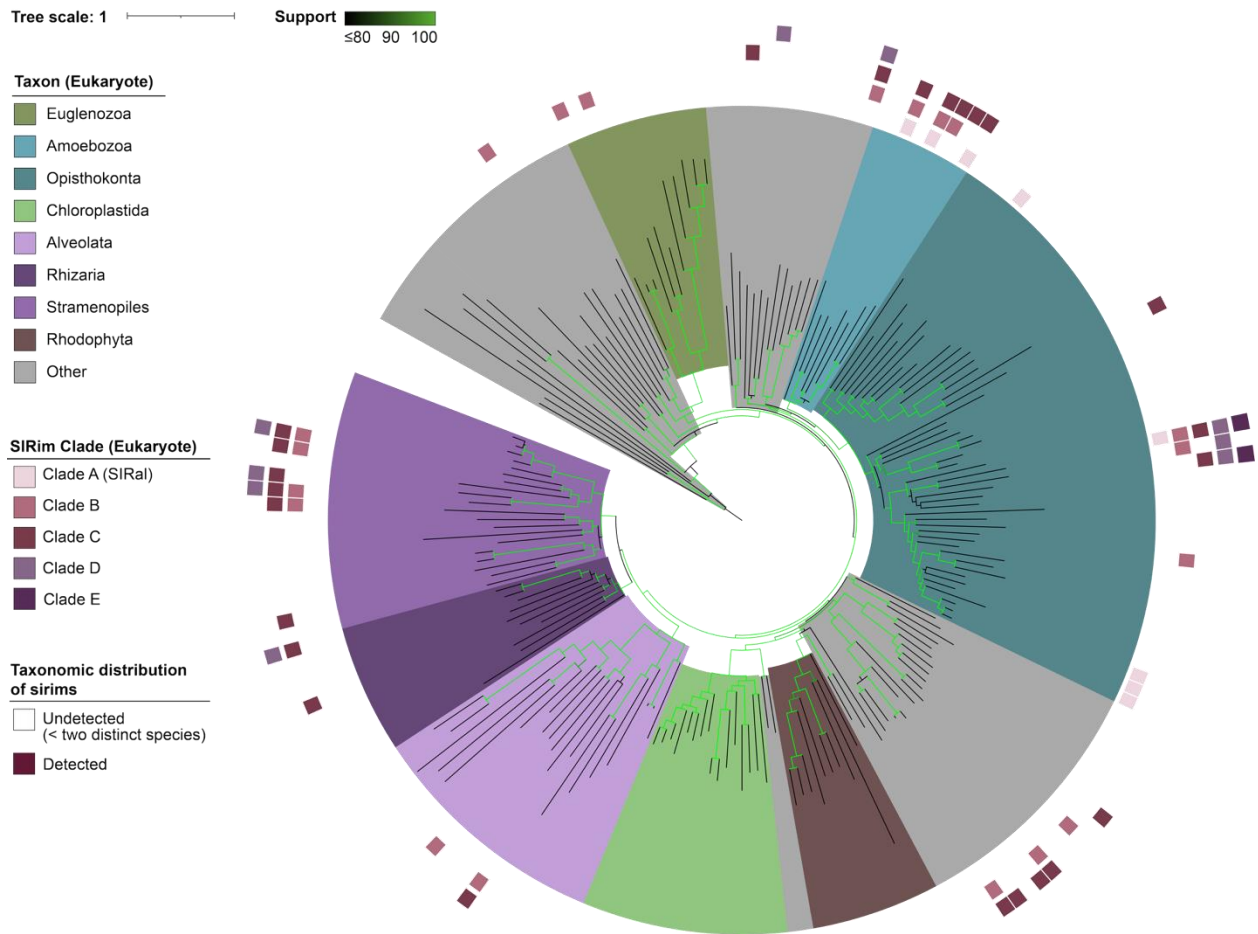
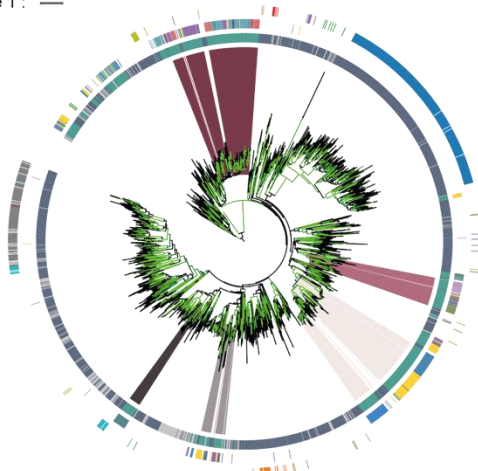


Fig. S5. Distribution of the five major eukaryotic SIRim clades identified in eukaryotes.

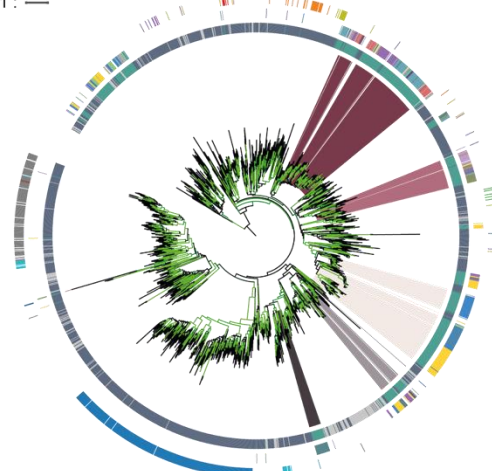
The phylogenetic tree from The Comparative Set (TCS) of Eukprot (196 species) is used to draw the taxonomic distribution of the SIRim proteins identified in the Eukprot v3 database of eukaryotic genomes (985 species) (39). SIRim proteins found in species present in the Eukprot v3 database (985 species) but not in the TCS (196 species) are assigned to their closest relative present in the TCS tree. We consider that a sirim protein is detected in the taxon represented by a given leaf of the tree if (i) sirims are detected in at least two distinct species represented by this leaf or (ii) the leaf represents a single eukaryotic species in Eukprot v3 and sirims are detected within this species. Eukaryotic sirims are grouped into five distinct clades based on our phylogenetic analysis (Fig. 4A, figs. S7 and S8). The tree is midpoint rooted. A pruned version of this tree can be found in Fig. 4A.

Tree scale 1 : —



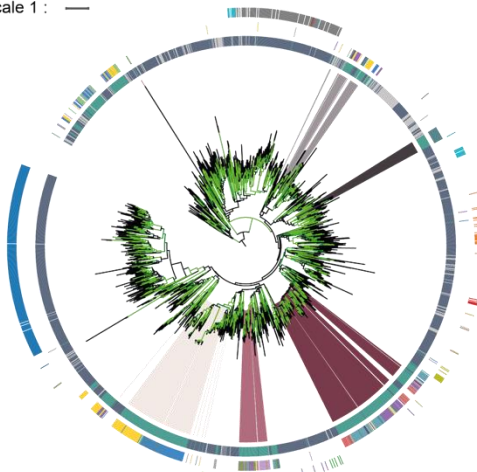
Muscle(none)

Tree scale 1 : —



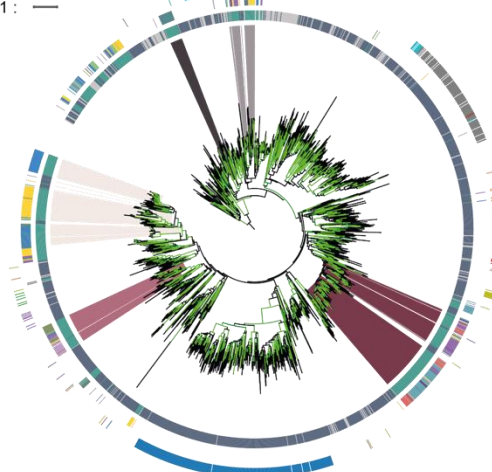
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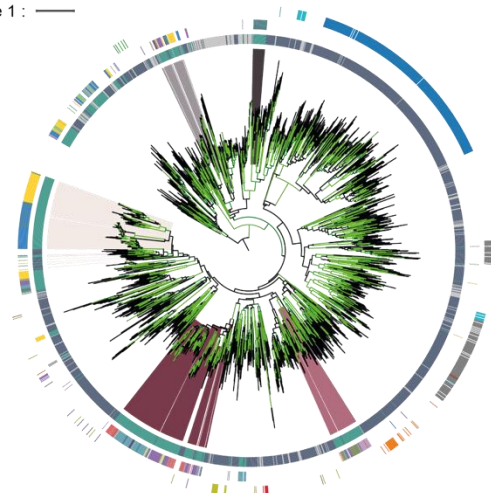
Muscle(acb)

Tree scale 1 : —



Muscle(bca)

Tree scale 1 : —



MAFFT L-INS-i

SIRim Clade (Eukaryote)

- Clade A (SIRal)
- Clade B
- Clade C
- Clade D
- Clade E

Domain of life

- Bacteria
- Archaea
- Eukaryote

Taxon (Eukaryote)

- Alveolata
- Stramenopiles
- Rhizaria
- Euglenozoa
- Haptophyta
- Chloroplastida
- Centroplasthelida
- Amoebozoa
- Opisthokonta
- Fungi
- Eumetazoa
- Other

Support ≤80 90 100

Antiphage system (Bacteria)

- pAgo
- PD-T7-2
- Lamassu
- Thoeris
- Dsr 1
- Dsr 2
- AbiH
- Avs
- Nezha
- Other

Fig. S6. The five major clades of eukaryotic sirims are stably identified regardless of the sequence alignment method.

The same representative sequences of bacterial, archaeal and eukaryotic sirims are used as in Fig. 4A. Representative sequences are aligned on their SIRim domain using either Muscle super5 (first four trees) or MAFFT (last tree, bottom). The Multiple Sequence Alignment (MSA) generated using Muscle are also built by permuting the guide tree (permutations: “none”, “abc”, “acb”, “bca”). The domain of life from which each protein originates (inner ring, blue, gray, green), the taxon for eukaryotic sirims (middle ring) and the corresponding antiphage system type for bacterial sirims (outer ring) are also indicated on each tree. One phylogenetic tree is built from each MSA using IQTree. This procedure allows us to test the stability of the identified eukaryotic clades of sirims (clades A to E) to perturbations of the MSA used to build the phylogenetic tree. We observed that the five clades of eukaryotic sirims defined from the phylogenetic tree in Fig. 4A are stably formed in all the obtained trees, indicating that they are significant. We also observe that the clades of eukaryotic sirims tend to associate with the same clades of bacterial antiphage sirims in all the trees.

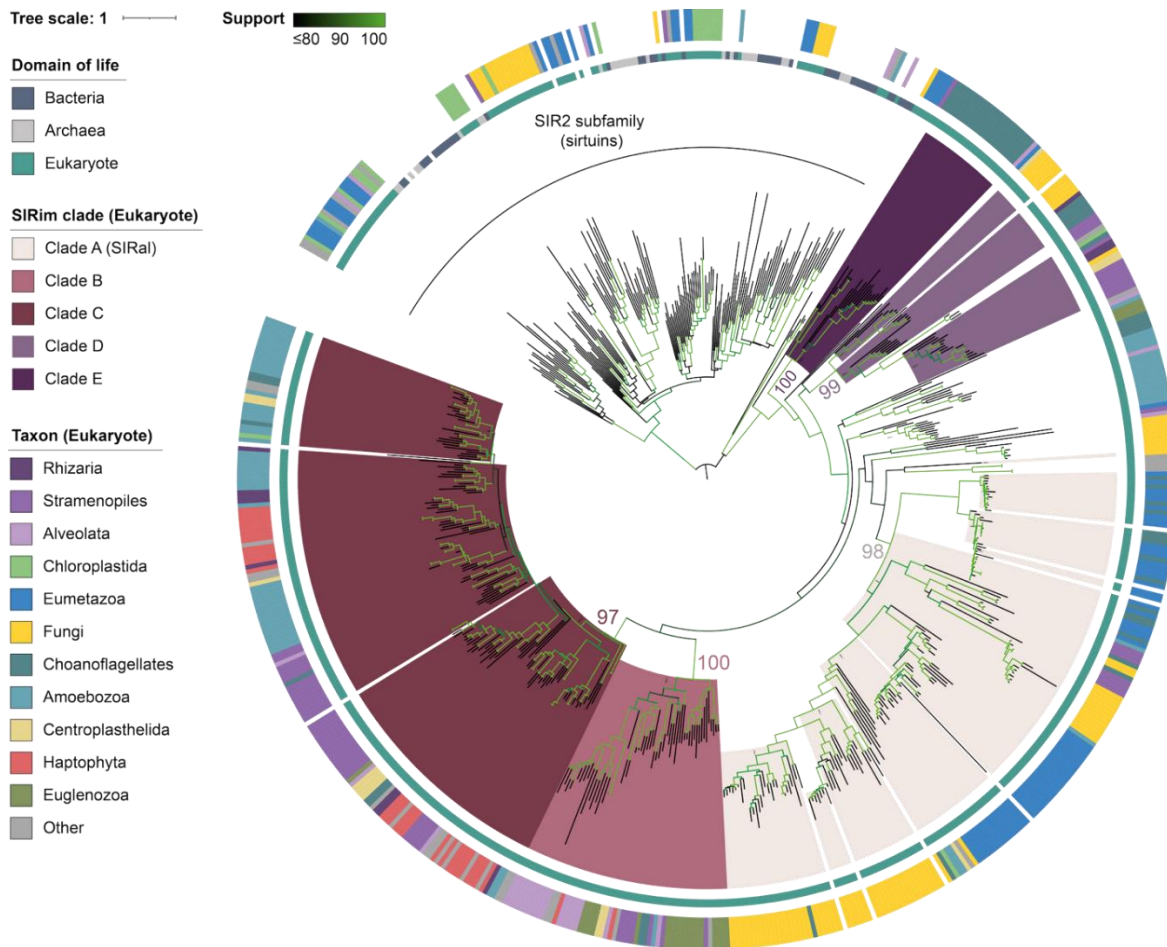


Fig. S7. Phylogenetic tree of the eukaryotic SIRim.

The tree only includes SIRim proteins from eukaryotic organisms and is rooted using the SIR2 subfamily of proteins (sirtuins) from across the tree of life as an outgroup. The five clades of eukaryotic sirims defined from the tree in Fig. 4A are represented (inner ring) and are also retrieved in this tree with good support. Eukaryotic sirim clades left in white branch are unstable when rebuilding the trees, so are left apart from the classification of eukaryotic sirims. The domain of life (middle ring) and the eukaryotic taxon (outer ring) are also indicated. Shades of green on branches indicate UltraFast Bootstrap support values. Support values are explicitly provided for key branches defining the major eukaryotic sirim clades.

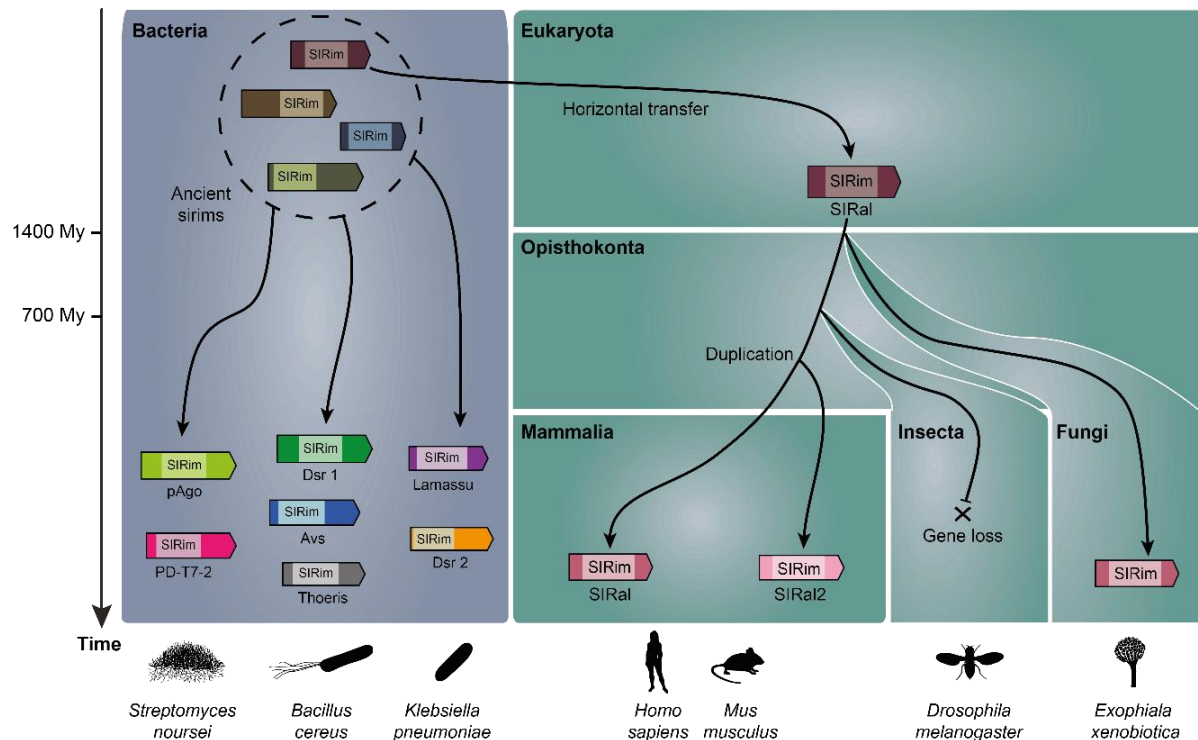


Fig. S8. Proposed evolutionary scenario explaining the apparition of the SIRal lineage through horizontal gene transfer from bacteria to eukaryotes and its subsequent diversification.

The evolutionary scenario presented above exemplifies three eukaryotic taxa: mammals, insects and fungi, even though SIRal lineage proteins are found in other taxa. For each taxon, example organisms are provided at the bottom of the figure. The depicted time scale is arbitrary.

In bacteria, the set of extant SIRim proteins present today originates from a pool of ancient SIRim proteins. In eukaryotes, homology detection and phylogenetic trees reveal several independent lineages of SIRim proteins, including the SIRal lineage (Clade A, Fig. 4A, figs. S5 to S7). This lineage is present in diverse eukaryotes including fungi and animals (opisthokonta), suggesting that it was present in eukaryotes prior to the last common ancestor of opisthokonta more than 1,400 million years ago (Clade A, Fig. 4A, figs. S5 to S7). Based on the trees, the ancestor to all SIRal lineage proteins was probably acquired from an ancient bacterial sirim through horizontal gene transfer, and then diversified with the apparition of the diverse eukaryotic taxa. In some taxa of opisthokonta, the gene was lost (e.g. in insects). In the rest of the taxa, the gene was maintained and can be found in extant biological organisms. In the genome of some animals such as mammals, two copies of the gene can be found (*SIRal*, originally named *FAM118B*, and *SIRal2*, named *FAM118A*). This suggests that the *SIRal* gene underwent a duplication event that led to two paralogs.

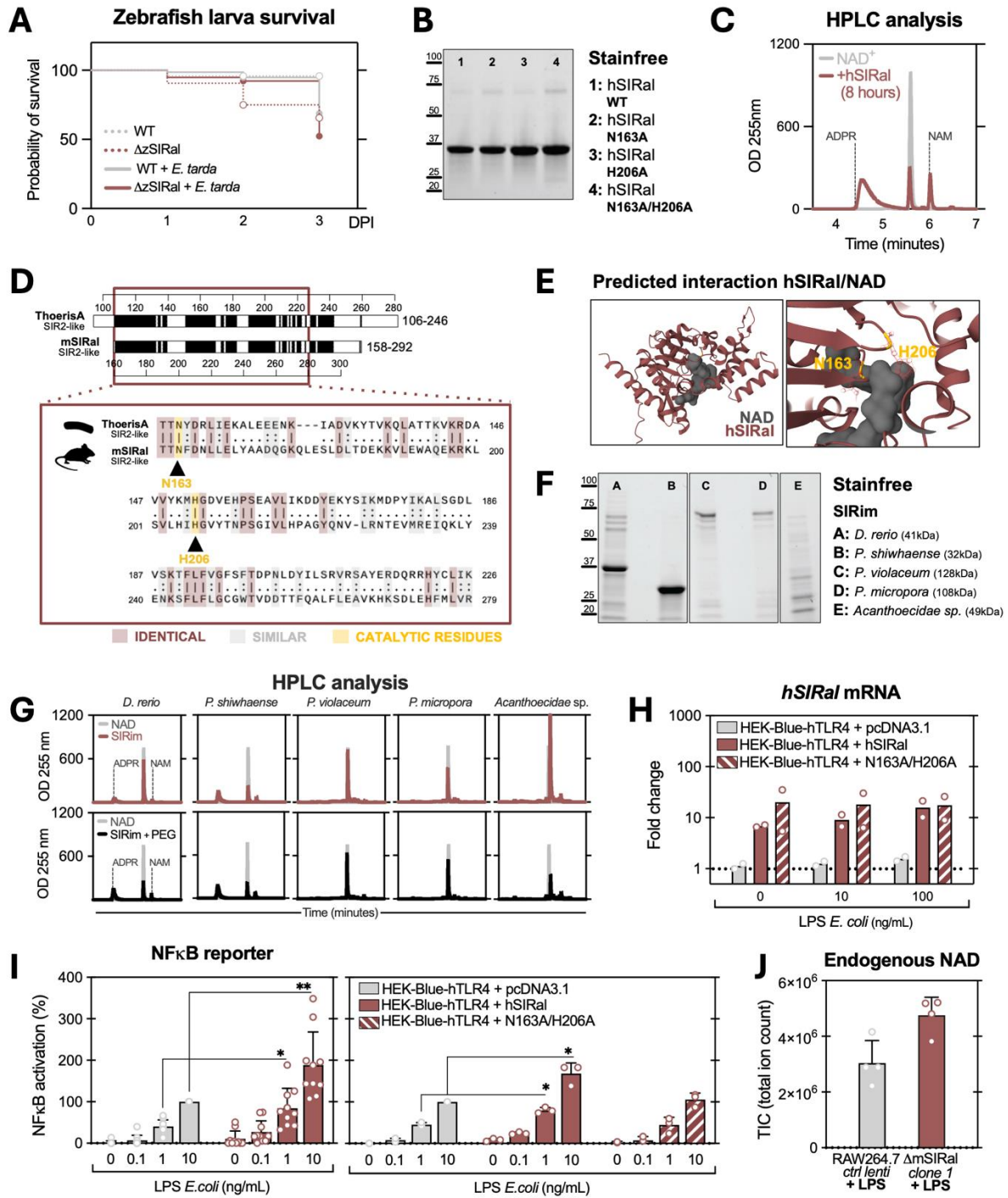


Fig. S9. Characterization of the NADase activities of SIRal and various Sirims.

(A) Wild type or $\Delta zSIRal$ zebrafish larvae were infected with *Edwardsiella tarda* and survival was monitored up to 3 days post-infection. **(B)** SDS-PAGE of recombinant purified hSIRal WT, N163A mutant, H206A mutant or N163A/H206A double-mutant. **(C)** High performance liquid chromatography (HPLC) analysis of NAD^+ with or without PEG after a 16h incubation with recombinant purified hSIRal. Standard elution volumes for ADPR and NAM are indicated. **(D)** Local alignment of the amino acid sequences of the SIRim domains of hSIRal and bacterial ThsA. Identical and similar residues are highlighted (red and grey, respectively), as well as the catalytic residues of ThsA (from (12), yellow). **(E)** AlphaFold prediction of hSIRal (red) interacting with NAD^+ (grey), with predicted interacting residues N163 and H206 (yellow). **(F)** SDS-PAGE of recombinant purified zSIRal from *D. rerio* (clade A, 41 kDa) or sirims from *P. shiwhaense* (clade B, 32 kDa), *P. violaceum* (clade C, 128 kDa), *P. micropora* (clade D, 108 kDa) or *Acanthoecidae* sp. (clade E, 49 kDa). **(G)** HPLC analysis of NAD^+ with or without 8h incubation with recombinant purified zSIRal from *D. rerio* (clade A) or sirims from *P. shiwhaense* (clade B), *P. violaceum* (clade C), *P. micropora* (clade D) or *Acanthoecidae* sp. (clade E). Standard elution volumes for ADPR and NAM are indicated. **(H, I)** HEK293T-hTLR4 NF- κ B reporter cells were transiently transfected with plasmids coding hSIRal WT or hSIRal N163A/H206A double-mutant and were stimulated with LPS from *E. coli* (0,1-10ng/mL). Levels of *hSIRal* transcripts were measured by RT-qPCR and NF- κ B activation was measured at 24h. Each dot represents the mean of technical replicates for each independent experiment. Bars correspond to the mean and error bars to the standard deviation of all the independent experiments. **(J)** Wild type or $\Delta mSIRal$ RAW264.7 cells were stimulated with LPS from *E. coli* (100ng/mL). Endogenous NAD^+ levels were analyzed by mass spectrometry at 4h.

Table S1.

Initial detection of proteins of the SIR2 family.

Table S2.

Curated detection of proteins of the SIR2 subfamily.

Table S3.

Curated detection of proteins of the SIRim subfamily.

Table S4.

Detection of sirims in the Eukprot v3 database of eukaryotic genomes.

TYPE		SEQUENCE
sgRNA	forward	CACCGGAAGTTATGGTAAGTAGTAC
sgRNA	reverse	AAACGTACTACTTACCATAACTTCC
Primer	forward	AAGCGGAAGCTAAGTGTGCTAC
Primer	reverse	AGAGACAACGAAGGAAGGTCTG

Table S5.
Primers and sgRNA used to construct *ΔmSIRal* / *FAM118B* RAW264.7 cells.

SPECIES	TARGET	REFERENCE
Mouse	<i>GAPDH</i>	Mm99999915_g15
Mouse	<i>FAM118B</i>	Mm00615099_m1
Mouse	<i>TNF</i>	Mm00443258_m1
Mouse	<i>iNOS (Nos2)</i>	Mm00440502_m1
Mouse	<i>Rsad2</i>	Mm00491265_m1
Human	<i>HPRT</i>	Hs02800695_m1
Human	<i>FAM118B</i>	Hs01090503_m1
Human	<i>IL8</i>	Hs00174103_m1
Human	<i>IL6</i>	Hs00174131_m1
Human	<i>TNF</i>	Hs00174128_m1
Human	<i>IFI44L</i>	Hs00915292_m1
Human	<i>Rsad2</i>	Hs00369813_m1
Human	<i>Mx1</i>	Hs00895608_m1

Table S6.

References of Taqman primers and probes used for RT-qPCR on RAW264.7 cells, mouse BMDMs and human monocyte-derived macrophages.

TARGET	FORWARD	REVERSE
<i>2F1A</i>	ACCTACCCTCCTCTTGGTCG	GGAACGGTGTGATTGAGGGAA
<i>IL22</i>	TGCAGAATCACTGTAAACACGA	CTCCCCGATTGCTTTGTTAC
<i>IL26</i>	ACTGCCAACTCAGTTCATTGAC	TTGCCTTCAGTCGCACAGTT
<i>IL1β</i>	GAGACAGACGGTGCTGTTTA	GTAAGACGGCAGTGAATCCA
<i>IFNγ</i>	CACATACAGTAGGATATTCACTCGC	TGGAAAGTAGCTTTCTTGCAGC
<i>IFNϕ1</i>	AATGGCTTGGCCGATACAGG	CACCTGGTCCTCCACCTTG

Table S7.

Sequences of primers used for RT-qPCR on zebrafish larvae.

References and Notes

1. F. Rousset, Innate immunity: The bacterial connection. *Trends Immunol.* **44**, 945–953 (2023). [doi:10.1016/j.it.2023.10.001](https://doi.org/10.1016/j.it.2023.10.001) [Medline](#)
2. B. R. Morehouse, Phage defense origin of animal immunity. *Curr. Opin. Microbiol.* **73**, 102295 (2023). [doi:10.1016/j.mib.2023.102295](https://doi.org/10.1016/j.mib.2023.102295) [Medline](#)
3. H. E. Ledvina, A. T. Whiteley, Conservation and similarity of bacterial and eukaryotic innate immunity. *Nat. Rev. Microbiol.* **22**, 420–434 (2024). [doi:10.1038/s41579-024-01017-1](https://doi.org/10.1038/s41579-024-01017-1) [Medline](#)
4. A. T. Whiteley, J. B. Eaglesham, C. C. de Oliveira Mann, B. R. Morehouse, B. Lowey, E. A. Nieminen, O. Danilchanka, D. S. King, A. S. Y. Lee, J. J. Mekalanos, P. J. Kranzusch, Bacterial cGAS-like enzymes synthesize diverse nucleotide signals. *Nature* **567**, 194–199 (2019). [doi:10.1038/s41586-019-0953-5](https://doi.org/10.1038/s41586-019-0953-5) [Medline](#)
5. B. R. Morehouse, A. A. Govande, A. Millman, A. F. A. Keszei, B. Lowey, G. Ofir, S. Shao, R. Sorek, P. J. Kranzusch, STING cyclic dinucleotide sensing originated in bacteria. *Nature* **586**, 429–433 (2020). [doi:10.1038/s41586-020-2719-5](https://doi.org/10.1038/s41586-020-2719-5) [Medline](#)
6. A. G. Johnson, T. Wein, M. L. Mayer, B. Duncan-Lowey, E. Yirmiya, Y. Oppenheimer-Shaanan, G. Amitai, R. Sorek, P. J. Kranzusch, Bacterial gasdermins reveal an ancient mechanism of cell death. *Science* **375**, 221–225 (2022). [doi:10.1126/science.abj8432](https://doi.org/10.1126/science.abj8432) [Medline](#)
7. T. Wein, A. G. Johnson, A. Millman, K. Lange, E. Yirmiya, R. Hadary, J. Garb, F. Steinruecke, A. B. Hill, P. J. Kranzusch, R. Sorek, CARD-like domains mediate anti-phage defense in bacterial gasdermin systems. *bioRxiv* 2023.05.28.542683 [Preprint] (2023); <https://doi.org/10.1101/2023.05.28.542683>.
8. A. Bernheim, A. Millman, G. Ofir, G. Meitav, C. Avraham, H. Shomar, M. M. Rosenberg, N. Tal, S. Melamed, G. Amitai, R. Sorek, Prokaryotic viperins produce diverse antiviral molecules. *Nature* **589**, 120–124 (2021). [doi:10.1038/s41586-020-2762-2](https://doi.org/10.1038/s41586-020-2762-2) [Medline](#)
9. H. Shomar, H. Georjon, Y. Feng, B. Olympio, M. Guillaume, F. Tesson, J. Cury, F. Wu, A. Bernheim, Viperin immunity evolved across the tree of life through serial innovations on a conserved scaffold. *Nat. Ecol. Evol.* **8**, 1667–1679 (2024). [doi:10.1038/s41559-024-02463-z](https://doi.org/10.1038/s41559-024-02463-z) [Medline](#)
10. B. R. Morehouse, M. C. J. Yip, A. F. A. Keszei, N. K. McNamara-Bordewick, S. Shao, P. J. Kranzusch, Cryo-EM structure of an active bacterial TIR-STING filament complex. *Nature* **608**, 803–807 (2022). [doi:10.1038/s41586-022-04999-1](https://doi.org/10.1038/s41586-022-04999-1) [Medline](#)
11. N. A. Lind, V. E. Rael, K. Pestal, B. Liu, G. M. Barton, Regulation of the nucleic acid-sensing Toll-like receptors. *Nat. Rev. Immunol.* **22**, 224–235 (2022). [doi:10.1038/s41577-021-00577-0](https://doi.org/10.1038/s41577-021-00577-0) [Medline](#)
12. G. Ofir, E. Herbst, M. Baroz, D. Cohen, A. Millman, S. Doron, N. Tal, D. B. A. Malheiro, S. Malitsky, G. Amitai, R. Sorek, Antiviral activity of bacterial TIR domains via immune signalling molecules. *Nature* **600**, 116–120 (2021). [doi:10.1038/s41586-021-04098-7](https://doi.org/10.1038/s41586-021-04098-7) [Medline](#)

13. A. M. Bayless, S. Chen, S. C. Ogden, X. Xu, J. D. Sidda, M. K. Manik, S. Li, B. Kobe, T. Ve, L. Song, M. Grant, L. Wan, M. T. Nishimura, Plant and prokaryotic TIR domains generate distinct cyclic ADPR NADase products. *Sci. Adv.* **9**, eade8487 (2023). [doi:10.1126/sciadv.ade8487](https://doi.org/10.1126/sciadv.ade8487) [Medline](#)
14. K. Essuman, J. Milbrandt, J. L. Dangl, M. T. Nishimura, Shared TIR enzymatic functions regulate cell death and immunity across the tree of life. *Science* **377**, eabo0001 (2022). [doi:10.1126/science.abo0001](https://doi.org/10.1126/science.abo0001) [Medline](#)
15. Q.-J. Wu, T.-N. Zhang, H.-H. Chen, X.-F. Yu, J.-L. Lv, Y.-Y. Liu, Y.-S. Liu, G. Zheng, J.-Q. Zhao, Y.-F. Wei, J.-Y. Guo, F.-H. Liu, Q. Chang, Y.-X. Zhang, C.-G. Liu, Y.-H. Zhao, The sirtuin family in health and disease. *Signal Transduct. Target. Ther.* **7**, 402 (2022). [doi:10.1038/s41392-022-01257-8](https://doi.org/10.1038/s41392-022-01257-8) [Medline](#)
16. K. Zhao, X. Chai, R. Marmorstein, Structure and substrate binding properties of cobB, a Sir2 homolog protein deacetylase from *Escherichia coli*. *J. Mol. Biol.* **337**, 731–741 (2004). [doi:10.1016/j.jmb.2004.01.060](https://doi.org/10.1016/j.jmb.2004.01.060) [Medline](#)
17. J. Garb, A. Lopatina, A. Bernheim, M. Zaremba, V. Siksnys, S. Melamed, A. Leavitt, A. Millman, G. Amitai, R. Sorek, Multiple phage resistance systems inhibit infection via SIR2-dependent NAD⁺ depletion. *Nat. Microbiol.* **7**, 1849–1856 (2022). [doi:10.1038/s41564-022-01207-8](https://doi.org/10.1038/s41564-022-01207-8) [Medline](#)
18. A. Millman, S. Melamed, A. Leavitt, S. Doron, A. Bernheim, J. Hör, J. Garb, N. Bechon, A. Brandis, A. Lopatina, G. Ofir, D. Hochhauser, A. Stokar-Avihail, N. Tal, S. Sharir, M. Voichek, Z. Erez, J. L. M. Ferrer, D. Dar, A. Kacen, G. Amitai, R. Sorek, An expanded arsenal of immune systems that protect bacteria from phages. *Cell Host Microbe* **30**, 1556–1569.e5 (2022). [doi:10.1016/j.chom.2022.09.017](https://doi.org/10.1016/j.chom.2022.09.017) [Medline](#)
19. D. Ka, H. Oh, E. Park, J.-H. Kim, E. Bae, Structural and functional evidence of bacterial antiphage protection by Thoeris defense system via NAD⁺ degradation. *Nat. Commun.* **11**, 2816 (2020). [doi:10.1038/s41467-020-16703-w](https://doi.org/10.1038/s41467-020-16703-w) [Medline](#)
20. G. Tamulaitiene, D. Sabonis, G. Sasnauskas, A. Ruksenaite, A. Silanskas, C. Avraham, G. Ofir, R. Sorek, M. Zaremba, V. Siksnys, Activation of Thoeris antiviral system via SIR2 effector filament assembly. *Nature* **627**, 431–436 (2024). [doi:10.1038/s41586-024-07092-x](https://doi.org/10.1038/s41586-024-07092-x) [Medline](#)
21. F. Tesson, A. Hervé, E. Mordret, M. Touchon, C. d’Humières, J. Cury, A. Bernheim, Systematic and quantitative view of the antiviral arsenal of prokaryotes. *Nat. Commun.* **13**, 2561 (2022). [doi:10.1038/s41467-022-30269-9](https://doi.org/10.1038/s41467-022-30269-9) [Medline](#)
22. D. A. Gold, D. A. Sinclair, Sirtuin evolution at the dawn of animal life. *Mol. Biol. Evol.* **39**, msac192 (2022). [doi:10.1093/molbev/msac192](https://doi.org/10.1093/molbev/msac192) [Medline](#)
23. S. Doron, S. Melamed, G. Ofir, A. Leavitt, A. Lopatina, M. Keren, G. Amitai, R. Sorek, Systematic discovery of antiphage defense systems in the microbial pangenome. *Science* **359**, eaar4120 (2018). [doi:10.1126/science.aar4120](https://doi.org/10.1126/science.aar4120) [Medline](#)
24. L. Gao, H. Altae-Tran, F. Böhning, K. S. Makarova, M. Segel, J. L. Schmid-Burgk, J. Koob, Y. I. Wolf, E. V. Koonin, F. Zhang, Diverse enzymatic activities mediate antiviral immunity in prokaryotes. *Science* **369**, 1077–1084 (2020). [doi:10.1126/science.aba0372](https://doi.org/10.1126/science.aba0372) [Medline](#)

25. K. E. Medvedev, L. N. Kinch, R. Dustin Schaeffer, J. Pei, N. V. Grishin, A fifth of the protein world: Rossmann-like proteins as an evolutionarily successful structural unit. *J. Mol. Biol.* **433**, 166788 (2021). [doi:10.1016/j.jmb.2020.166788](https://doi.org/10.1016/j.jmb.2020.166788) [Medline](#)
26. H. Yuan, R. Marmorstein, Structural basis for sirtuin activity and inhibition. *J. Biol. Chem.* **287**, 42428–42435 (2012). [doi:10.1074/jbc.R112.372300](https://doi.org/10.1074/jbc.R112.372300) [Medline](#)
27. J. C. Opazo, M. W. Vandewege, F. G. Hoffmann, K. Zavala, C. Meléndez, C. Luchsinger, V. A. Cavieres, L. Vargas-Chacoff, F. J. Morera, P. V. Burgos, C. Tapia-Rojas, G. A. Mardones, How many sirtuin genes are out there? Evolution of sirtuin genes in vertebrates with a description of a new family member. *Mol. Biol. Evol.* **40**, msad014 (2023). [doi:10.1093/molbev/msad014](https://doi.org/10.1093/molbev/msad014) [Medline](#)
28. Y. Li, K. Fong, M. Tang, X. Han, Z. Gong, W. Ma, M. Hebert, Z. Songyang, J. Chen, Fam118B, a novel component in Cajal bodies, is required for Cajal body formation, snRNP biogenesis and cell viability. *J. Cell Sci.* **127**, 2029–2039 (2014). [doi:10.1242/jcs.143453](https://doi.org/10.1242/jcs.143453) [Medline](#)
29. L. G. Gardinassi, A cross-study biomarker signature of human bronchial epithelial cells infected with respiratory syncytial virus. *Adv. Virol.* **2016**, 3605302 (2016). [doi:10.1155/2016/3605302](https://doi.org/10.1155/2016/3605302) [Medline](#)
30. J. Cury, M. Haudiquet, V. Hernandez Trejo, E. Mordret, A. Hanouna, M. Rotival, F. Tesson, D. Bonhomme, G. Ofir, L. Quintana-Murci, P. Benaroch, E. Z. Poirier, A. Bernheim, Conservation of antiviral systems across domains of life reveals immune genes in humans. *Cell Host Microbe* **32**, 1594–1607.e5 (2024). [doi:10.1016/j.chom.2024.08.002](https://doi.org/10.1016/j.chom.2024.08.002) [Medline](#)
31. M. Yamamoto, S. Sato, H. Hemmi, H. Sanjo, S. Uematsu, T. Kaisho, K. Hoshino, O. Takeuchi, M. Kobayashi, T. Fujita, K. Takeda, S. Akira, Essential role for TIRAP in activation of the signalling cascade shared by TLR2 and TLR4. *Nature* **420**, 324–329 (2002). [doi:10.1038/nature01182](https://doi.org/10.1038/nature01182) [Medline](#)
32. T. Ve, P. R. Vajjhala, A. Hedger, T. Croll, F. DiMaio, S. Horsefield, X. Yu, P. Lavrencic, Z. Hassan, G. P. Morgan, A. Mansell, M. Mobli, A. O’Carroll, B. Chauvin, Y. Gambin, E. Sieriecki, M. J. Landsberg, K. J. Stacey, E. H. Egelman, B. Kobe, Structural basis of TIR-domain-assembly formation in MAL- and MyD88-dependent TLR4 signaling. *Nat. Struct. Mol. Biol.* **24**, 743–751 (2017). [doi:10.1038/nsmb.3444](https://doi.org/10.1038/nsmb.3444) [Medline](#)
33. J. Shi, Y. Zhao, Y. Wang, W. Gao, J. Ding, P. Li, L. Hu, F. Shao, Inflammatory caspases are innate immune receptors for intracellular LPS. *Nature* **514**, 187–192 (2014). [doi:10.1038/nature13683](https://doi.org/10.1038/nature13683) [Medline](#)
34. M. Yamamoto, S. Sato, H. Hemmi, K. Hoshino, T. Kaisho, H. Sanjo, O. Takeuchi, M. Sugiyama, M. Okabe, K. Takeda, S. Akira, Role of adaptor TRIF in the MyD88-independent toll-like receptor signaling pathway. *Science* **301**, 640–643 (2003). [doi:10.1126/science.1087262](https://doi.org/10.1126/science.1087262) [Medline](#)
35. F. Heil, H. Hemmi, H. Hochrein, F. Ampenberger, C. Kirschning, S. Akira, G. Lipford, H. Wagner, S. Bauer, Species-specific recognition of single-stranded RNA via toll-like receptor 7 and 8. *Science* **303**, 1526–1529 (2004). [doi:10.1126/science.1093620](https://doi.org/10.1126/science.1093620) [Medline](#)
36. K. Honda, H. Yanai, T. Mizutani, H. Negishi, N. Shimada, N. Suzuki, Y. Ohba, A. Takaoka, W.-C. Yeh, T. Taniguchi, Role of a transductional-transcriptional processor complex

- involving MyD88 and IRF-7 in Toll-like receptor signaling. *Proc. Natl. Acad. Sci. U.S.A.* **101**, 15416–15421 (2004). [doi:10.1073/pnas.0406933101](https://doi.org/10.1073/pnas.0406933101) [Medline](#)
37. A. Ablasser, S. Hur, Regulation of cGAS- and RLR-mediated immunity to nucleic acids. *Nat. Immunol.* **21**, 17–29 (2020). [doi:10.1038/s41590-019-0556-1](https://doi.org/10.1038/s41590-019-0556-1) [Medline](#)
 38. M. A. Anwar, S. Panneerselvam, M. Shah, S. Choi, Insights into the species-specific TLR4 signaling mechanism in response to *Rhodobacter sphaeroides* lipid A detection. *Sci. Rep.* **5**, 7657 (2015). [doi:10.1038/srep07657](https://doi.org/10.1038/srep07657) [Medline](#)
 39. D. J. Richter, C. Berney, J. F. H. Strassert, Y.-P. Poh, E. K. Herman, S. A. Muñoz-Gómez, J. G. Wideman, F. Burki, C. De Vargas, EukProt: A database of genome-scale predicted proteins across the diversity of eukaryotes. *Peer Community J.* **2**, e56 (2022). [doi:10.24072/pcjournal.173](https://doi.org/10.24072/pcjournal.173)
 40. E. M. Kibby, A. N. Conte, A. M. Burroughs, T. A. Nagy, J. A. Vargas, L. A. Whalen, L. Aravind, A. T. Whiteley, Bacterial NLR-related proteins protect against phage. *Cell* **186**, 2410–2424.e18 (2023). [doi:10.1016/j.cell.2023.04.015](https://doi.org/10.1016/j.cell.2023.04.015) [Medline](#)
 41. F. Rousset, E. Yirmiya, S. Neshet, A. Brandis, T. Mehlman, M. Itkin, S. Malitsky, A. Millman, S. Melamed, R. Sorek, A conserved family of immune effectors cleaves cellular ATP upon viral infection. *Cell* **186**, 3619–3631.e13 (2023). [doi:10.1016/j.cell.2023.07.020](https://doi.org/10.1016/j.cell.2023.07.020) [Medline](#)
 42. P. Broz, V. M. Dixit, Inflammasomes: Mechanism of assembly, regulation and signalling. *Nat. Rev. Immunol.* **16**, 407–420 (2016). [doi:10.1038/nri.2016.58](https://doi.org/10.1038/nri.2016.58) [Medline](#)
 43. J. Heller, C. Clavé, P. Gladieux, S. J. Saupe, N. L. Glass, NLR surveillance of essential SEC-9 SNARE proteins induces programmed cell death upon allorecognition in filamentous fungi. *Proc. Natl. Acad. Sci. U.S.A.* **115**, E2292–E2301 (2018). [doi:10.1073/pnas.1719705115](https://doi.org/10.1073/pnas.1719705115) [Medline](#)
 44. Y. Shi, V. Masic, T. Mosaiab, P. Rajaratman, L. Hartley-Tassell, M. Sorbello, C. C. Goulart, E. Vasquez, B. P. Mishra, S. Holt, W. Gu, B. Kobe, T. Ve, Structural characterization of macro domain-containing Thoeris antiphage defense systems. *Sci. Adv.* **10**, eadn3310 (2024). [doi:10.1126/sciadv.adn3310](https://doi.org/10.1126/sciadv.adn3310) [Medline](#)
 45. D. Sabonis, C. Avraham, R. B. Chang, A. Lu, E. Herbst, A. Silanskas, D. Vilutis, A. Leavitt, E. Yirmiya, H. C. Toyoda, A. Ruksenaite, M. Zaremba, I. Osterman, G. Amitai, P. J. Kranzusch, R. Sorek, G. Tamulaitiene, TIR domains produce histidine-ADPR conjugates as an immune signal molecules in bacteria. *Nature* **642**, 467–473 (2025). [doi:10.1038/s41586-025-08930-2](https://doi.org/10.1038/s41586-025-08930-2) [Medline](#)
 46. Y. Li, Y. Li, X. Cao, X. Jin, T. Jin, Pattern recognition receptors in zebrafish provide functional and evolutionary insight into innate immune signaling pathways. *Cell. Mol. Immunol.* **14**, 80–89 (2017). [doi:10.1038/cmi.2016.50](https://doi.org/10.1038/cmi.2016.50) [Medline](#)
 47. O. Parnas, M. Jovanovic, T. M. Eisenhaure, R. H. Herbst, A. Dixit, C. J. Ye, D. Przybylski, R. J. Platt, I. Tirosh, N. E. Sanjana, O. Shalem, R. Satija, R. Raychowdhury, P. Mertins, S. A. Carr, F. Zhang, N. Hacohen, A. Regev, A genome-wide CRISPR screen in primary immune cells to dissect regulatory networks. *Cell* **162**, 675–686 (2015). [doi:10.1016/j.cell.2015.06.059](https://doi.org/10.1016/j.cell.2015.06.059) [Medline](#)

48. D. Fisch, T. Zhang, H. Sun, W. Ma, Y. Tan, S. P. Gygi, D. E. Higgins, J. C. Kagan, Molecular definition of the endogenous Toll-like receptor signalling pathways. *Nature* **631**, 635–644 (2024). [doi:10.1038/s41586-024-07614-7](https://doi.org/10.1038/s41586-024-07614-7) [Medline](#)
49. Y. Li, K. M. Slavik, H. C. Toyoda, B. R. Morehouse, C. C. De Oliveira Mann, A. Elek, S. Levy, Z. Wang, K. S. Mears, J. Liu, D. Kashin, X. Guo, T. Mass, A. Seb  -Pedr  s, F. Schwede, P. J. Kranzusch, cGLRs are a diverse family of pattern recognition receptors in innate immunity. *Cells* **186**, 3261–3276.e20 (2023). [doi:10.1016/j.cell.2023.05.038](https://doi.org/10.1016/j.cell.2023.05.038) [Medline](#)
50. H. Cai, L. Li, K. M. Slavik, J. Huang, T. Yin, X. Ai, L. H  delin, G. Haas, Z. Xiang, Y. Yang, X. Li, Y. Chen, Z. Wei, H. Deng, D. Chen, R. Jiao, N. Martins, C. Meignin, P. J. Kranzusch, J.-L. Imler, The virus-induced cyclic dinucleotide 2'3'-c-di-GMP mediates STING-dependent antiviral immunity in *Drosophila*. *Immunity* **56**, 1991–2005.e9 (2023). [doi:10.1016/j.immuni.2023.08.006](https://doi.org/10.1016/j.immuni.2023.08.006) [Medline](#)
51. K. M. Slavik, B. R. Morehouse, A. E. Ragucci, W. Zhou, X. Ai, Y. Chen, L. Li, Z. Wei, H. B  hre, M. K  nig, R. Seifert, A. S. Y. Lee, H. Cai, J.-L. Imler, P. J. Kranzusch, cGAS-like receptors sense RNA and control 3'2'-cGAMP signalling in *Drosophila*. *Nature* **597**, 109–113 (2021). [doi:10.1038/s41586-021-03743-5](https://doi.org/10.1038/s41586-021-03743-5) [Medline](#)
52. A. Bernheim, J. Cury, E. Z. Poirier, The immune modules conserved across the tree of life: Towards a definition of ancestral immunity. *PLOS Biol.* **22**, e3002717 (2024). [doi:10.1371/journal.pbio.3002717](https://doi.org/10.1371/journal.pbio.3002717) [Medline](#)
53. E. W. Sayers, M. Cavanaugh, K. Clark, J. Ostell, K. D. Pruitt, I. Karsch-Mizrachi, GenBank. *Nucleic Acids Res.* **47**, D94–D99 (2019). [doi:10.1093/nar/gky989](https://doi.org/10.1093/nar/gky989) [Medline](#)
54. D. H. Parks, M. Chuvochina, C. Rinke, A. J. Mussig, P.-A. Chaumeil, P. Hugenholtz, GTDB: An ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy. *Nucleic Acids Res.* **50**, D785–D794 (2022). [doi:10.1093/nar/gkab776](https://doi.org/10.1093/nar/gkab776) [Medline](#)
55. R. C. Edgar, Muscle5: High-accuracy alignment ensembles enable unbiased assessments of sequence homology and phylogeny. *Nat. Commun.* **13**, 6968 (2022). [doi:10.1038/s41467-022-34630-w](https://doi.org/10.1038/s41467-022-34630-w) [Medline](#)
56. S. R. Eddy, Accelerated profile HMM searches. *PLOS Comput. Biol.* **7**, e1002195 (2011). [doi:10.1371/journal.pcbi.1002195](https://doi.org/10.1371/journal.pcbi.1002195) [Medline](#)
57. M. Hauser, M. Steinegger, J. S  ding, MMseqs software suite for fast and deep clustering and searching of large protein sequence sets. *Bioinformatics* **32**, 1323–1330 (2016). [doi:10.1093/bioinformatics/btw006](https://doi.org/10.1093/bioinformatics/btw006) [Medline](#)
58. B. Q. Minh, H. A. Schmidt, O. Chernomor, D. Schrempf, M. D. Woodhams, A. von Haeseler, R. Lanfear, IQ-TREE 2: New models and efficient methods for phylogenetic inference in the genomic era. *Mol. Biol. Evol.* **37**, 1530–1534 (2020). [doi:10.1093/molbev/msaa015](https://doi.org/10.1093/molbev/msaa015) [Medline](#)
59. M. Steinegger, J. S  ding, MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat. Biotechnol.* **35**, 1026–1028 (2017). [doi:10.1038/nbt.3988](https://doi.org/10.1038/nbt.3988) [Medline](#)

60. P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Í. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt; SciPy 1.0 Contributors, SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nat. Methods* **17**, 261–272 (2020). [doi:10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2) [Medline](#)
61. J. L. Steenwyk, T. J. Buida III, Y. Li, X.-X. Shen, A. Rokas, ClipKIT: A multiple sequence alignment trimming software for accurate phylogenomic inference. *PLOS Biol.* **18**, e3001007 (2020). [doi:10.1371/journal.pbio.3001007](https://doi.org/10.1371/journal.pbio.3001007) [Medline](#)
62. S. Kalyaanamoorthy, B. Q. Minh, T. K. F. Wong, A. von Haeseler, L. S. Jermini, ModelFinder: Fast model selection for accurate phylogenetic estimates. *Nat. Methods* **14**, 587–589 (2017). [doi:10.1038/nmeth.4285](https://doi.org/10.1038/nmeth.4285) [Medline](#)
63. D. T. Hoang, O. Chernomor, A. von Haeseler, B. Q. Minh, L. S. Vinh, UFBoot2: Improving the ultrafast bootstrap approximation. *Mol. Biol. Evol.* **35**, 518–522 (2018). [doi:10.1093/molbev/msx281](https://doi.org/10.1093/molbev/msx281) [Medline](#)
64. B. Buchfink, K. Reuter, H.-G. Drost, Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nat. Methods* **18**, 366–368 (2021). [doi:10.1038/s41592-021-01101-x](https://doi.org/10.1038/s41592-021-01101-x) [Medline](#)
65. K. Katoh, D. M. Standley, MAFFT multiple sequence alignment software version 7: Improvements in performance and usability. *Mol. Biol. Evol.* **30**, 772–780 (2013). [doi:10.1093/molbev/mst010](https://doi.org/10.1093/molbev/mst010) [Medline](#)
66. E. M. Culbertson, T. C. Levin, Eukaryotic CD-NTase, STING, and viperin proteins evolved via domain shuffling, horizontal transfer, and ancient inheritance from prokaryotes. *PLOS Biol.* **21**, e3002436 (2023). [doi:10.1371/journal.pbio.3002436](https://doi.org/10.1371/journal.pbio.3002436) [Medline](#)
67. W. Shen, H. Ren, TaxonKit: A practical and efficient NCBI taxonomy toolkit. *J. Genet. Genomics* **48**, 844–850 (2021). [doi:10.1016/j.jgg.2021.03.006](https://doi.org/10.1016/j.jgg.2021.03.006) [Medline](#)
68. S. Kumar, M. Suleski, J. M. Craig, A. E. Kasprowitz, M. Sanderford, M. Li, G. Stecher, S. B. Hedges, TimeTree 5: An expanded resource for species divergence times. *Mol. Biol. Evol.* **39**, msac174 (2022). [doi:10.1093/molbev/msac174](https://doi.org/10.1093/molbev/msac174) [Medline](#)
69. F. Gabler, S. Z. Nam, S. Till, M. Mirdita, M. Steinegger, J. Söding, A. N. Lupas, V. Alva, Protein sequence analysis using the MPI bioinformatics toolkit. *Curr. Protoc. Bioinformatics* **72**, e108 (2020). [doi:10.1002/cpbi.108](https://doi.org/10.1002/cpbi.108) [Medline](#)
70. M. Mirdita, K. Schütze, Y. Moriwaki, L. Heo, S. Ovchinnikov, M. Steinegger, ColabFold: Making protein folding accessible to all. *Nat. Methods* **19**, 679–682 (2022). [doi:10.1038/s41592-022-01488-1](https://doi.org/10.1038/s41592-022-01488-1) [Medline](#)
71. J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S. A. A. Kohl, A. J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A. W. Senior, K. Kavukcuoglu, P.

- Kohli, D. Hassabis, Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021). [doi:10.1038/s41586-021-03819-2](https://doi.org/10.1038/s41586-021-03819-2) [Medline](#)
72. A. N. Shah, C. F. Davey, A. C. Whitebirch, A. C. Miller, C. B. Moens, Rapid reverse genetic screening using CRISPR in zebrafish. *Nat. Methods* **12**, 535–540 (2015). [doi:10.1038/nmeth.3360](https://doi.org/10.1038/nmeth.3360)